

FDU 回归分析 期中考试 (2023 秋)

Total: 100 marks

Duration: 2 hour and 30 minutes

Problem 1

Consider the simple linear regression model $y_i = \alpha + \beta x_i + \varepsilon_i$ ($i = 1, 2, \dots, n$) where $\{\varepsilon_i\}_{i=1}^n \stackrel{iid}{\sim} N(0, \sigma^2)$.

Denote
$$\begin{cases} \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \\ s^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta} x_i)^2 \end{cases} \quad \text{where } (\hat{\alpha}, \hat{\beta}) \text{ are LSE of parameters } (\alpha, \beta)$$

(1) Show that $\bar{y} - E(\bar{y})$ is independent of $\hat{\beta} - \beta$ [5 Marks]

(2) Show that $\bar{y}$ is independent of s^2 [5 Marks]

(3) Find the confidence interval (with confidence level $1 - \tau$) of $\alpha + \beta x$, simultaneously for all x . [15 Marks]

Preparation

我们记:

$$\begin{cases} \gamma = [\alpha, \beta]^T \\ \varepsilon = [\varepsilon_1, \dots, \varepsilon_n]^T \\ x = [x_1, \dots, x_n]^T \\ y = [y_1, \dots, y_n]^T \\ X = [1_n, x] \in \mathbb{R}^{n \times 2} \end{cases}$$

则我们有:

$$\begin{aligned} y &= X\gamma + \varepsilon \\ \hat{\gamma} &= (X^T X)^{-1} X^T y \\ &= \begin{bmatrix} 1_n^T 1_n & 1_n^T x \\ 1_n^T x & x^T x \end{bmatrix}^{-1} \begin{bmatrix} 1_n^T y \\ x^T y \end{bmatrix} \\ &= \frac{1}{nx^T x - (1_n^T x)^2} \begin{bmatrix} x^T x & -1_n^T x \\ -1_n^T x & n \end{bmatrix} \begin{bmatrix} 1_n^T y \\ x^T y \end{bmatrix} \Rightarrow \begin{cases} \hat{\alpha} = \bar{y} - \bar{x} \hat{\beta} = \bar{y} - \bar{x} \frac{S_{xy}}{S_{xx}} \\ \hat{\beta} = \frac{S_{xy}}{S_{xx}} \end{cases} \\ &= \frac{1}{n(x^T x - n\bar{x}^2)} \begin{bmatrix} x^T x \cdot n\bar{y} - x^T y \cdot n\bar{x} \\ n(x^T y - n\bar{x}\bar{y}) \end{bmatrix} \\ &= \begin{bmatrix} \bar{y} - \bar{x} \frac{S_{xy}}{S_{xx}} \\ \frac{S_{xy}}{S_{xx}} \end{bmatrix} \end{aligned}$$

Part (1)

Show that $\bar{y} - E(\bar{y})$ is independent of $\hat{\beta} - \beta$ [5 Marks]

Solution:

我们有:

$$\begin{aligned}
\hat{\beta} &= \frac{S_{xy}}{S_{xx}} \\
&= \frac{x^T y - n\bar{x}\bar{y}}{S_{xx}} \\
&= \frac{(x - \bar{x}1_n)^T y}{S_{xx}} \\
&= \frac{(x - \bar{x}1_n)^T (\alpha 1_n + \beta x + \varepsilon)}{S_{xx}} \\
&= \frac{1}{S_{xx}} \{ \alpha (x^T 1_n - \bar{x} 1_n^T 1_n) + \beta (x^T x - \bar{x} 1_n^T x) + (x - \bar{x} 1_n)^T \varepsilon \} \\
&= \frac{1}{S_{xx}} \{ \alpha (n\bar{x} - n\bar{x}) + \beta (x^T x - n\bar{x}^2) + (x - \bar{x} 1_n)^T \varepsilon \} \\
&= \frac{1}{S_{xx}} \{ \alpha \cdot 0 + \beta \cdot S_{xx} + (x - \bar{x} 1_n)^T \varepsilon \} \\
&= \beta + \frac{(x - \bar{x} 1_n)^T \varepsilon}{S_{xx}} \\
\hline
\bar{y} &= \frac{1}{n} 1_n^T y \\
&= \frac{1}{n} 1_n^T (\alpha 1_n + \beta x + \varepsilon) \\
&= \alpha + \beta \bar{x} + \frac{1}{n} 1_n^T \varepsilon \\
&= E[\bar{y}] + \frac{1}{n} 1_n^T \varepsilon
\end{aligned}$$

在正态假设下，要证明 $\bar{y} - E[\bar{y}]$ 与 $\hat{\beta} - \beta$ 独立，等价于证明它们不相关：

$$\begin{aligned}
\text{Cov}(\bar{y} - E[\bar{y}], \hat{\beta} - \beta) &= \text{Cov} \left(\frac{1}{n} 1_n^T \varepsilon, \frac{(x - \bar{x} 1_n)^T \varepsilon}{S_{xx}} \right) \\
&= \frac{1}{n} 1_n^T \text{Cov}(\varepsilon, \varepsilon) \frac{(x - \bar{x} 1_n)}{S_{xx}} \\
&= \frac{1}{n} 1_n^T \cdot \sigma^2 I_n \cdot \frac{(x - \bar{x} 1_n)}{S_{xx}} \\
&= \frac{\sigma^2}{S_{xx}} \left(\frac{1}{n} 1_n^T x - \frac{1}{n} 1_n^T 1_n \cdot \bar{x} \right) \\
&= \frac{\sigma^2}{S_{xx}} (\bar{x} - \bar{x}) \\
&= 0
\end{aligned}$$

因此 $\bar{y} - E[\bar{y}]$ 与 $\hat{\beta} - \beta$ 相互独立。

Part (2)

Show that $\bar{y}$ is independent of s^2 [5 Marks]

Solution:

$$\begin{aligned}
s^2 &= \frac{1}{n-2} \|y - \hat{y}\|_2^2 \\
&= \frac{1}{n-2} \|y - Hy\|_2^2 \quad (H = X(X^T X)^{-1} X^T) \\
&= \frac{1}{n-2} y^T (I_n - H)^T (I_n - H) y \\
&= \frac{1}{n-2} y^T (I_n - H) y \\
&= \frac{1}{n-2} (X\gamma + \varepsilon)^T (I_n - H) (X\gamma + \varepsilon) \\
&= \frac{1}{n-2} \varepsilon^T (I_n - H) \varepsilon \\
\hline
\bar{y} &= \frac{1}{n} 1_n^T y \\
&= \frac{1}{n} 1_n^T (\alpha 1_n + \beta x + \varepsilon) \\
&= \alpha + \beta \bar{x} + \frac{1}{n} 1_n^T \varepsilon \\
&= E[\bar{y}] + \frac{1}{n} 1_n^T \varepsilon
\end{aligned}$$

要证明 $\bar{y}$ 与 s^2 独立, 只需证明 $\frac{1}{n} 1_n^T \varepsilon$ 与 $(I_n - H)\varepsilon$ 独立, 在正态假设下, 即证明它们不相关:

$$\begin{aligned}
\text{Cov}\left(\frac{1}{n} 1_n^T \varepsilon, (I_n - H)\varepsilon\right) &= \frac{1}{n} 1_n^T \text{Cov}(\varepsilon, \varepsilon) (I_n - H) \\
&= \frac{1}{n} 1_n^T \cdot \sigma^2 I_n \cdot (I_n - H) \\
&= \frac{\sigma^2}{n} 1_n^T (I_n - H) \\
&= \frac{\sigma^2}{n} (1_n^T - 1_n^T) \\
&= 0
\end{aligned}$$

因此 $\bar{y}$ 与 s^2 独立.

Part (3)

Find the confidence interval (with confidence level $1 - \tau$) of $\alpha + \beta x$, simultaneously for all x . [15 Marks]

Solution:

$$\begin{aligned}
E[\hat{\gamma}] &= \gamma \\
\hline
\text{Cov}(\hat{\gamma}) &= \sigma^2 (X^T X)^{-1} \\
&= \sigma^2 \begin{bmatrix} 1_n^T 1_n & 1_n^T x \\ 1_n^T x & x^T x \end{bmatrix}^{-1} \\
&= \sigma^2 \frac{1}{nx^T x - (1_n^T x)^2} \begin{bmatrix} x^T x & -1_n^T x \\ -1_n^T x & n \end{bmatrix} \\
&= \sigma^2 \frac{1}{nS_{xx}} \begin{bmatrix} x^T x & -n\bar{x} \\ -n\bar{x} & n \end{bmatrix} \\
&= \sigma^2 \begin{bmatrix} \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} & -\frac{\bar{x}}{S_{xx}} \\ -\frac{\bar{x}}{S_{xx}} & \frac{1}{S_{xx}} \end{bmatrix}
\end{aligned}$$

因此我们有:

$$\begin{aligned} \frac{\mathbb{E}[\hat{\alpha} + \hat{\beta}x] = \alpha + \beta x}{\text{Var}[\hat{\alpha} + \hat{\beta}x] &= \text{Var}(\hat{\alpha}) + 2x\text{Cov}(\hat{\alpha}, \hat{\beta}) + x^2\text{Var}(\hat{\beta})} \\ &= \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\sigma^2 + 2x \cdot \left(-\frac{\bar{x}\sigma^2}{S_{xx}}\right) + x^2 \frac{\sigma^2}{S_{xx}} \\ &= \sigma^2 \left(\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}\right) \end{aligned}$$

上述问题等价于寻找一个常数 M_α 使得:

$$\mathbb{P} \left\{ \frac{[(\hat{\alpha} + \hat{\beta}x) - (\alpha + \beta x)]^2}{s^2[\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}]} \leq M_\tau^2 \text{ for all } x \right\} = 1 - \tau$$

即等价于使得:

$$\mathbb{P} \left\{ \max_x \frac{[(\hat{\alpha} + \hat{\beta}x) - (\alpha + \beta x)]^2}{s^2[\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}]} \leq M_\tau^2 \right\} = 1 - \tau$$

(Statistical Inference 习题 11.40)

若 a, b, c, d 为常数且 $c, d > 0$, 则 $\max_t \frac{(a+bt)^2}{c+dt^2} = \frac{a^2}{c} + \frac{b^2}{d}$

这个结论是下面引理的直接推论.

Lemma: (广义 Rayleigh 商)

若 $b \in \mathbb{R}^n$ 为给定向量且 $A \in \mathbb{R}^{n \times n}$ 正定, 则 $\max_{x \neq 0, n \in \mathbb{R}^n} \frac{(b^T x)^2}{x^T A x} = b^T A^{-1} b$

证明:

$$\begin{aligned} \max_{x \neq 0, n \in \mathbb{R}^n} \frac{(b^T x)^2}{x^T A x} &= \max_{y \neq 0, n \in \mathbb{R}^n} \frac{(b^T A^{-\frac{1}{2}} y)^2}{y^T y} \quad (y := A^{\frac{1}{2}} x) \\ &= \max_{y \neq 0, n \in \mathbb{R}^n} \frac{y^T (A^{-\frac{1}{2}} b b^T A^{-\frac{1}{2}}) y}{y^T y} \quad (\text{Rayleigh theorem}) \\ &= \lambda_{\max}(A^{-\frac{1}{2}} b b^T A^{-\frac{1}{2}}) \quad (\text{note that } A \text{ is positive definite, hence symmetric}) \\ &= \lambda_{\max}\{(A^{-\frac{1}{2}} b)(A^{-\frac{1}{2}} b)^T\} \quad (\text{note that rank-one matrix } zz^T \text{'s only non-zero eigenvalue is } \|z\|_2^2) \\ &= \|A^{-\frac{1}{2}} b\|_2^2 \\ &= b^T A^{-1} b \end{aligned}$$

其中最大值可以在 $x = \alpha A^{-1} b$ ($\forall \alpha \neq 0 \in \mathbb{R}$) 取到.

$$\begin{aligned}
\max_x \frac{[(\hat{\alpha} + \hat{\beta}x) - (\alpha + \beta x)]^2}{s^2 \left[\frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}} \right]} &= \frac{1}{s^2} \max_x \frac{[(\bar{y} - \hat{\beta}\bar{x} + \hat{\beta}x) - (\alpha + \beta\bar{x} - \beta\bar{x} + \beta x)]^2}{\frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}} \\
&= \frac{1}{s^2} \max_x \frac{[(\bar{y} - \alpha - \beta\bar{x}) + (\hat{\beta} - \beta)(x - \bar{x})]^2}{\frac{1}{n} + \frac{1}{S_{xx}}(x - \bar{x})^2} \quad (\text{denote } t = x - \bar{x}) \\
&= \frac{1}{s^2} \max_t \frac{[(\bar{y} - \alpha - \beta\bar{x}) + (\hat{\beta} - \beta)t]^2}{\frac{1}{n} + \frac{1}{S_{xx}}t^2} \quad (\text{use lemma}) \\
&= \frac{1}{s^2} \left\{ \frac{(\bar{y} - \alpha - \beta\bar{x})^2}{\frac{1}{n}} + \frac{(\hat{\beta} - \beta)^2}{\frac{1}{S_{xx}}} \right\} \\
&= \frac{\frac{(\bar{y} - \alpha - \beta\bar{x})^2}{\sigma^2/n} + \frac{(\hat{\beta} - \beta)^2}{\sigma^2/S_{xx}}}{s^2/\sigma^2} \\
&\quad (\text{note that } \begin{cases} \bar{y} \sim N(\alpha + \beta\bar{x}, \frac{\sigma^2}{n}) \\ \hat{\beta} \sim N(\beta, \frac{\sigma^2}{S_{xx}}) \\ s^2 \sim \frac{1}{n-2} \sigma^2 \chi_{(n-2)}^2 \end{cases} \text{ and they are independent}) \\
&\sim \frac{\chi_{(2)}^2}{\chi_{(n-2)}^2/(n-2)} \\
&= 2 \frac{\chi_{(2)}^2/2}{\chi_{(n-2)}^2/(n-2)} \\
&= 2F_{2,n-2}
\end{aligned}$$

因此 $M_\tau^2 = 2F_{2,n-2,\tau}$, 即 $M_\tau = \sqrt{2F_{2,n-2,\tau}}$

于是我们得到 $\alpha + \beta x$ ($\forall x$) 的 $1 - \tau$ 置信区间为 $\hat{\alpha} + \hat{\beta}x \pm \sqrt{s^2 \left[\frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}} \right] \cdot 2F_{2,n-2,\tau}}$

Problem 2

Consider the linear regression model $y = X\beta + \varepsilon$,

where $\begin{cases} E(\varepsilon) = 0_n \\ \text{Cov}(\varepsilon) = \sigma^2 \Sigma \end{cases}$ in which Σ is a known positive definite matrix.

Find the BLUE for β and derive its variance-covariance matrix. [10 Marks]

Solution:

由于 $\Sigma \in \mathbb{R}^{n \times n}$ 是已知的正定矩阵, 因此平方根 $\Sigma^{\frac{1}{2}}$ 及其逆矩阵 $\Sigma^{-\frac{1}{2}}$ 都是存在的
于是我们有:

$$\begin{aligned}
y &= X\beta + \varepsilon \\
&\Leftrightarrow \\
\Sigma^{-\frac{1}{2}}y &= \Sigma^{-\frac{1}{2}}X\beta + \Sigma^{-\frac{1}{2}}\varepsilon
\end{aligned}$$

定义:

$$\begin{cases} \tilde{y} = \Sigma^{-\frac{1}{2}}y \\ \tilde{X} = \Sigma^{-\frac{1}{2}}X \Rightarrow \tilde{y} = \tilde{X}\beta + \tilde{\varepsilon} \\ \tilde{\varepsilon} = \Sigma^{-\frac{1}{2}}\varepsilon \end{cases}$$

则我们有:

$$\begin{aligned}
\mathbf{E}[\tilde{\varepsilon}] &= \mathbf{E}[\Sigma^{-\frac{1}{2}} \varepsilon] \\
&= \Sigma^{-\frac{1}{2}} \mathbf{E}[\varepsilon] \\
&= \Sigma^{-\frac{1}{2}} \mathbf{0}_n \\
&= \mathbf{0}_n \\
\hline
\text{Cov}[\tilde{\varepsilon}] &= \text{Cov}[\Sigma^{-\frac{1}{2}} \varepsilon] \\
&= \Sigma^{-\frac{1}{2}} \text{Cov}[\varepsilon] \Sigma^{-\frac{1}{2}} \\
&= \Sigma^{-\frac{1}{2}} \Sigma \Sigma^{-\frac{1}{2}} \\
&= \mathbf{I}_n
\end{aligned}$$

根据 Gauss-Markov 定理可知 β 的 BLUE 即为新模型下的 LSE:

$$\begin{aligned}
\hat{\beta} &:= (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T y \\
&= (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} y
\end{aligned}$$

其协方差矩阵为:

$$\begin{aligned}
\text{Cov}(\hat{\beta}) &= \text{Cov}((X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} y) \\
&= (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} \cdot \text{Cov}(y) \cdot \Sigma^{-1} X (X^T \Sigma^{-1} X)^{-1} \\
&= (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} \cdot \text{Cov}(\varepsilon) \cdot \Sigma^{-1} X (X^T \Sigma^{-1} X)^{-1} \\
&= (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} \cdot \sigma^2 \Sigma \cdot \Sigma^{-1} X (X^T \Sigma^{-1} X)^{-1} \\
&= \sigma^2 (X^T \Sigma^{-1} X)^{-1}
\end{aligned}$$

Problem 3

Consider a multiple linear regression model with p independent variables $y = X\beta + \varepsilon$ where $\varepsilon \sim N(0, \sigma^2 I_n)$

Denote $\begin{cases} \hat{\beta} = (X^T X)^{-1} X^T y \\ s^2 = \frac{1}{n-p-1} \|y - X\hat{\beta}\|^2 \end{cases}$

(1) Find $\text{Var}(s^2)$ [5 Marks]

(2) Let $A = \frac{1}{n-p-3} (I_n - XX^T)$, Find $\mathbf{E}[(y^T A y - \sigma^2)^2]$ [10 Marks]

(3) In terms of mean square errors, which estimators of σ^2 is better, $y^T A y$ or s^2 ? [5 Marks]

Part (1)

Find $\text{Var}(s^2)$ [5 Marks]

Solution:

Gamma 随机变量 $W \sim \text{Gamma}(\alpha, \lambda)$ 的均值和方差分别为:

$$\begin{cases} \mathbf{E}[W] = \frac{\alpha}{\lambda} \\ \text{Var}[W] = \frac{\alpha}{\lambda^2} \end{cases}$$

考虑 s^2 的分布:

$$\begin{aligned}
s^2 &= \frac{1}{n-p-1} \|y - X\hat{\beta}\|_2^2 \quad (\text{recall that } \hat{y} = X\hat{\beta} = Hy \text{ where } H = X(X^T X)^{-1} X^T) \\
&= \frac{1}{n-p-1} \|y - Hy\|_2^2 \\
&= \frac{1}{n-p-1} y^T (I_n - H)^T (I_n - H) y \quad (\text{note that } H \text{ is symmetric and idempotent}) \\
&= \frac{1}{n-p-1} y^T (I_n - H) y \quad (\text{note that } y = X\beta + \varepsilon) \\
&= \frac{1}{n-p-1} (X\beta + \varepsilon)^T (I_n - H) (X\beta + \varepsilon) \quad (\text{note that } HX = X \text{ so that } (I_n - H)X = 0_{n \times (p+1)}) \\
&= \frac{1}{n-p-1} \varepsilon^T (I_n - H) \varepsilon \quad (\text{note that } \text{tr}(H) = p+1 \Rightarrow \text{tr}(I_n - H) = n-p-1) \\
&\sim \frac{1}{n-p-1} \sigma^2 \chi_{n-p-1}^2
\end{aligned}$$

因此其方差为:

$$\begin{aligned}
\text{Var}(s^2) &= \text{Var}\left[\frac{1}{n-p-1} \sigma^2 \chi_{n-p-1}^2\right] \\
&= \left(\frac{1}{n-p-1} \sigma^2\right)^2 \cdot \text{Var}[\chi_{n-p-1}^2] \\
&= \frac{1}{(n-p-1)^2} \sigma^4 \cdot \text{Var}\left[\text{Gamma}\left(\frac{n-p-1}{2}, \frac{1}{2}\right)\right] \\
&= \frac{1}{(n-p-1)^2} \sigma^4 \cdot \frac{\frac{n-p-1}{2}}{\left(\frac{1}{2}\right)^2} \\
&= \frac{2\sigma^4}{n-p-1}
\end{aligned}$$

Part (2)

Let $A = \frac{1}{n-p-3} (I_n - XX^\dagger)$, Find $E[(y^T A y - \sigma^2)^2]$ [10 Marks]

Solution:

在 $X \in \mathbb{R}^{n \times (p+1)}$ 列满秩的假设下, Moore-Penrose 逆 $X^\dagger = (X^T X)^{-1} X^T$

记 $H := XX^\dagger = X(X^T X)^{-1} X^T$

可以证明其对称、幂等且迹 $\text{tr}(H) := p+1$, 而且 $\text{Range}(H) = \text{Range}(XX^\dagger) = \text{Range}(X)$

则我们有:

$$\begin{aligned}
A &= \frac{1}{n-p-3} (I_n - XX^\dagger) = \frac{1}{n-p-3} (I_n - H) \\
\hline
y^T A y &= y^T \cdot \frac{1}{n-p-3} (I_n - H) \cdot y \\
&= \frac{1}{n-p-3} y^T (I_n - H) y \\
&= \frac{1}{n-p-3} (X\beta + \varepsilon)^T (I_n - H) (X\beta + \varepsilon) \quad (\text{note that } HX = X \Rightarrow (I_n - H)X = 0_{n \times (p+1)}) \\
&= \frac{1}{n-p-3} \varepsilon^T (I_n - H) \varepsilon \quad (H \text{ is symmetric and idempotent } \begin{cases} \text{tr}(I_n - H) = n-p-1 \\ \text{Cov}(\varepsilon) = \sigma^2 I_n \end{cases}) \\
&\sim \frac{1}{n-p-3} \sigma^2 \chi_{(n-p-1)}^2
\end{aligned}$$

因此我们有:

$$\begin{aligned}
E[y^T Ay] &= \frac{1}{n-p-3} \sigma^2 \cdot (n-p-1) = \frac{n-p-1}{n-p-3} \sigma^2 \\
\text{Var}[y^T Ay] &= \left(\frac{1}{n-p-3} \sigma^2 \right)^2 \cdot 2(n-p-1) = \frac{2(n-p-1)}{(n-p-3)^2} \sigma^4 \\
E[(y^T Ay)^2] &= \text{Var}[y^T Ay] + \{E[y^T Ay]\}^2 \\
&= \frac{2(n-p-1)}{(n-p-3)^2} \sigma^4 + \left(\frac{n-p-1}{n-p-3} \sigma^2 \right)^2 \\
&= \frac{(n-p-1)(n-p+1)}{(n-p-3)^2} \sigma^4
\end{aligned}$$

于是我们有:

$$\begin{aligned}
E[(y^T Ay - \sigma^2)^2] &= E[(y^T Ay)^2 - 2\sigma^2 y^T Ay + \sigma^4] \\
&= E[(y^T Ay)^2] - 2\sigma^2 E[y^T Ay] + \sigma^4 \\
&= \frac{(n-p-1)(n-p+1)}{(n-p-3)^2} \sigma^4 - 2\sigma^2 \frac{n-p-1}{n-p-3} \sigma^2 + \sigma^4 \\
&= \left\{ \frac{(n-p-1)(n-p+1)}{(n-p-3)^2} - \frac{2(n-p-1)}{n-p-3} + 1 \right\} \sigma^4 \\
&= \frac{(n-p-1)(n-p+1) - 2(n-p-1)(n-p-3) + (n-p-3)^2}{(n-p-3)^2} \sigma^4 \\
&= \frac{-(n-p-1)(n-p-7) + (n-p-3)^2}{(n-p-3)^2} \sigma^4 \\
&= \frac{2(n-p+1)}{(n-p-3)^2} \sigma^4
\end{aligned}$$

或者更简单地:

$$\begin{aligned}
E[(y^T Ay - \sigma^2)^2] &= \text{Var}[y^T Ay - \sigma^2] + \{E[y^T Ay - \sigma^2]\}^2 \\
&= \text{Var}[y^T Ay] + \{E[y^T Ay] - \sigma^2\}^2 \\
&= \frac{2(n-p-1)}{(n-p-3)^2} \sigma^4 + \left\{ \frac{n-p-1}{n-p-3} \sigma^2 - \sigma^2 \right\}^2 \\
&= \frac{2(n-p-1)}{(n-p-3)^2} \sigma^4 + \left\{ \frac{2}{n-p-3} \sigma^2 \right\}^2 \\
&= \frac{2(n-p-1)}{(n-p-3)^2} \sigma^4 + \frac{4}{(n-p-3)^2} \sigma^4 \\
&= \frac{2(n-p+1)}{(n-p-3)^2} \sigma^4
\end{aligned}$$

Part (3)

In terms of mean square errors, which estimators of σ^2 is better, $y^T Ay$ or s^2 ? [5 Marks]

Solution:

根据第 (1) 问可知 $\text{Var}(s^2) = \frac{2\sigma^4}{n-p-1}$

注意到 $E[s^2] = E\left[\frac{1}{n-p-1} \sigma^2 \chi_{n-p-1}^2\right] = \frac{1}{n-p-1} \sigma^2 (n-p-1) = \sigma^2$

因此我们有:

$$\begin{aligned}
\text{MSE}(s^2) &= E[(s^2 - \sigma^2)^2] \quad (\text{note that } E[s^2] = \sigma^2) \\
&= E[(s^2 - E[s^2])^2] \\
&= \text{Var}(s^2) \\
&= \frac{2\sigma^4}{n-p-1}
\end{aligned}$$

根据第 (2) 问可知 $\text{MSE}(y^T Ay) = \mathbb{E}[(y^T Ay - \sigma^2)^2] = \frac{2(n-p+1)}{(n-p-3)^2} \sigma^4$

则我们有:

$$\begin{aligned}
 \text{MSE}(y^T Ay) - \text{MSE}(s^2) &= \frac{2(n-p+1)}{(n-p-3)^2} \sigma^4 - \frac{2}{n-p-1} \sigma^4 \\
 &= \frac{(n-p+1)(n-p-1) - (n-p-3)^2}{(n-p-3)^2(n-p-1)} \cdot (2\sigma^4) \\
 &= \frac{(n-p)^2 - 1 - (n-p)^2 + 6(n-p) - 9}{(n-p-3)^2(n-p-1)} \cdot (2\sigma^4) \\
 &= \frac{6(n-p) - 10}{(n-p-3)^2(n-p-1)} \cdot (2\sigma^4) \quad (\text{note that } n > k+1) \\
 &\geq \frac{6 \cdot 2 - 10}{(n-p-3)^2(n-p-1)} \cdot (2\sigma^4) \\
 &= \frac{2}{(n-p-3)^2(n-p-1)} \cdot (2\sigma^4) \\
 &> 0
 \end{aligned}$$

因此从均方误差 (Mean Square Error, MSE) 的角度来说, $y^T Ay$ 对 σ^2 的估计效果要比 s^2 差.

Problem 4

DEFINITION:

The **multiple correlation coefficient** between a random variable Y and a random vector $(X_1, X_2, \dots, X_k)^T$ is the maximum correlation between Y and any linear function $\alpha_1 X_1 + \dots + \alpha_k X_k$ of $(X_1, X_2, \dots, X_k)^T$.

Now consider a multiple linear regression model where its response variable Y and the k predictors $X_1, X_2, \dots, X_k$ are multivariate normal. Find the multiple correlation coefficient between Y and $X_1, X_2, \dots, X_k$.

[HINT: use the Cauchy-Schwarz inequality for $\|\cdot\|_2$: $|u^T v| \leq \|u\|_2 \cdot \|v\|_2$] [15 Marks]

Solution:

记 $X = [X_1, X_2, \dots, X_k]^T$

根据题意可知 Y, X 服从联合多元正态分布, 记为:

$$\begin{bmatrix} Y \\ X \end{bmatrix} \sim N \left(\begin{bmatrix} \mu_Y \\ \mu_X \end{bmatrix}, \begin{bmatrix} \sigma_{YY}^2 & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{bmatrix} \right)$$

其中 $\Sigma_{YX} = \Sigma_{XY}^T$ 且 Σ_{XX} 为对称阵.

记系数向量为 $\alpha = [\alpha_1, \alpha_2, \dots, \alpha_k]^T$

根据题干中的定义可知 Y 与 $X = [X_1, X_2, \dots, X_k]^T$ 之间的多重共线性系数 r 为:

$$\begin{aligned}
 r &:= \max_{\alpha \in \mathbb{R}^k} \text{Corr}(Y, \alpha^T X) \\
 &= \max_{\alpha \in \mathbb{R}^k} \frac{\text{Cov}(Y, \alpha^T X)}{\sqrt{\text{Var}(Y) \text{Var}(\alpha^T X)}} \\
 &= \max_{\alpha \in \mathbb{R}^k} \frac{\alpha^T \text{Cov}(Y, X)}{\sqrt{\text{Var}(Y) \alpha^T \text{Var}(X) \alpha}} \\
 &= \max_{\alpha \in \mathbb{R}^k} \frac{\alpha^T \Sigma_{XY}}{\sqrt{\sigma_{YY}^2 \alpha^T \Sigma_{XX} \alpha}}
 \end{aligned}$$

根据题干中的提示, 我们对 $\alpha^T \Sigma_{YX}$ 应用 Cauchy-Schwarz 不等式可得:

$$\begin{aligned}
\alpha^T \Sigma_{XY} &= \alpha^T \Sigma_{XX}^{\frac{1}{2}} \Sigma_{XX}^{-\frac{1}{2}} \Sigma_{XY} \\
&= (\Sigma_{XX}^{\frac{1}{2}} \alpha)^T (\Sigma_{XX}^{-\frac{1}{2}} \Sigma_{XY}) \quad (\text{Cauchy-Schwarz Inequality}) \\
&\leq \|\Sigma_{XX}^{\frac{1}{2}} \alpha\|_2 \|\Sigma_{XX}^{-\frac{1}{2}} \Sigma_{XY}\|_2 \\
&= \sqrt{\alpha^T \Sigma_{XX} \alpha} \cdot \sqrt{\Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}} \quad (\text{note that } \Sigma_{YX} = \Sigma_{XY}^T)
\end{aligned}$$

上述不等式当且仅当 $\Sigma_{XX}^{\frac{1}{2}} \alpha$ 和 $\Sigma_{XX}^{-\frac{1}{2}} \Sigma_{XY}$ 线性相关时取等。

于是我们有：

$$\begin{aligned}
r &= \max_{\alpha \in \mathbb{R}^k} \frac{\alpha^T \Sigma_{XY}}{\sqrt{\sigma_{YY}^2 \alpha^T \Sigma_{XX} \alpha}} \\
&= \frac{\sqrt{\alpha^T \Sigma_{XX} \alpha} \cdot \sqrt{\Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}}}{\sqrt{\sigma_{YY}^2 \alpha^T \Sigma_{XX} \alpha}} \\
&= \frac{\sqrt{\Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}}}{\sigma_{YY}}
\end{aligned}$$

根据多元线性回归模型的假设，我们有 $Y = \beta_0 + \beta^T X + \varepsilon$

其中 $\beta \in \mathbb{R}^k$ 且 $\varepsilon \sim N(0, \sigma^2)$ 与 X 独立。

$$\text{因此} \begin{bmatrix} Y \\ X \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta^T X + \varepsilon \\ X \end{bmatrix} = \begin{bmatrix} \beta_0 \\ 0_k \end{bmatrix} + \begin{bmatrix} 1 & \beta^T \\ 0_k & I_k \end{bmatrix} \begin{bmatrix} \varepsilon \\ X \end{bmatrix}$$

其协方差矩阵为：

$$\begin{aligned}
\text{Cov} \left(\begin{bmatrix} Y \\ X \end{bmatrix} \right) &= \text{Cov} \left(\begin{bmatrix} \beta_0 \\ 0_k \end{bmatrix} + \begin{bmatrix} 1 & \beta^T \\ 0_k & I_k \end{bmatrix} \begin{bmatrix} \varepsilon \\ X \end{bmatrix} \right) \\
&= \begin{bmatrix} 1 & \beta^T \\ 0_k & I_k \end{bmatrix} \text{Cov} \left(\begin{bmatrix} \varepsilon \\ X \end{bmatrix} \right) \begin{bmatrix} 1 & \beta^T \\ 0_k & I_k \end{bmatrix}^T \\
&= \begin{bmatrix} 1 & \beta^T \\ 0_k & I_k \end{bmatrix} \begin{bmatrix} \sigma^2 & \\ & \Sigma_{XX} \end{bmatrix} \begin{bmatrix} 1 & \beta^T \\ 0_k & I_k \end{bmatrix}^T \\
&= \begin{bmatrix} \sigma^2 & \beta^T \Sigma_{XX} \\ 0_k & \Sigma_{XX} \end{bmatrix} \begin{bmatrix} 1 & 0_k^T \\ \beta & I_k \end{bmatrix} \\
&= \begin{bmatrix} \sigma^2 + \beta^T \Sigma_{XX} \beta & \beta^T \Sigma_{XX} \\ \Sigma_{XX} \beta & \Sigma_{XX} \end{bmatrix}
\end{aligned}$$

因此我们有：

$$\begin{aligned}
\sigma_{YY}^2 &= \sigma^2 + \beta^T \Sigma_{XX} \beta \\
\Sigma_{XY} &= \Sigma_{XX} \beta \\
\Sigma_{YX} &= \beta^T \Sigma_{XX} \\
r &= \frac{\sqrt{\Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}}}{\sigma_{YY}} = \frac{\sqrt{\beta^T \Sigma_{XX} \Sigma_{XX}^{-1} \Sigma_{XX} \beta}}{\sqrt{\sigma^2 + \beta^T \Sigma_{XX} \beta}} = \sqrt{\frac{\beta^T \Sigma_{XX} \beta}{\sigma^2 + \beta^T \Sigma_{XX} \beta}}
\end{aligned}$$

Problem 5

Given the multiple linear regression model with k independent variables:

$$y = X\beta + \varepsilon \text{ where } \varepsilon \sim N(0, \sigma^2 I_n)$$

Consider the following hypothesis testing:

$$H_0 : C\beta = h \quad \Leftrightarrow \quad H_1 : C\beta \neq h$$

where $C \in \mathbb{R}^{m \times (k+1)}$ is a rank- m constant matrix and $h \in \mathbb{R}^m$ is a constant vector.

(1) Find the LSE (least squares estimator) of β under H_0 [10 marks]

(2) Find the F-test for testing H_0

(Write the test statistic and prove its null distribution) [10 marks]

Part (1)

Find the LSE (least squares estimator) of β under H_0 [10 marks]

Solution:

考虑求解线性约束最小二乘问题 $\min_{C\beta=h} \|y - X\beta\|_2^2$

注意到目标函数 $f(\beta) = \|y - X\beta\|_2^2$ 是关于 β 的凸函数，而问题只有线性等式约束 $C\beta = h$ 因此这是一个标准形式的凸优化问题，其最优解即为 KKT 点。

定义其 Lagrange 函数 $L(\beta, \lambda)$ 为:

$$\begin{aligned} L(\beta, \lambda) &= f(\beta) - \lambda^T (C\beta - h) \\ &= \|y - X\beta\|_2^2 - \lambda^T (C\beta - h) \end{aligned}$$
$$\text{dom}\{L\} = \mathbb{R}^{p+1} \times \mathbb{R}^m$$

Lagrange 函数 $L(\beta, \lambda)$ 关于 β 的梯度为:

$$\begin{aligned} \nabla_{\beta} L(\beta, \lambda) &= \nabla_{\beta} \{ \|y - X\beta\|_2^2 - \lambda^T (C\beta - h) \} \\ &= -X^T \cdot 2(y - X\beta) - (\lambda^T C)^T \\ &= -2X^T y + 2X^T X\beta - C^T \lambda \end{aligned}$$

KKT 条件为:

$$\begin{cases} \nabla_{\beta} L(\beta, \lambda) = -2X^T y + 2X^T X\beta - C^T \lambda = 0_{p+1} & \textcircled{1} \\ C\beta = h & \textcircled{2} \end{cases}$$

① 式左乘 $(X^T X)^{-1}$ 可得 $-2(X^T X)^{-1} X^T y + 2\beta - (X^T X)^{-1} C^T \lambda = 0_{p+1}$

于是有 $\beta = (X^T X)^{-1} X^T y + \frac{1}{2} (X^T X)^{-1} C^T \lambda$

代入 ② 式即得 $C\beta = C(X^T X)^{-1} X^T y + \frac{1}{2} C(X^T X)^{-1} C^T \lambda = h$

解得 $\lambda_{\text{KKT}} = 2[C(X^T X)^{-1} C^T]^{-1} [h - C(X^T X)^{-1} X^T y]$

因此我们有:

$$\begin{aligned} \hat{\beta}_{\text{reduced}} &= \beta_{\text{KKT}} \\ &= (X^T X)^{-1} X^T y + \frac{1}{2} (X^T X)^{-1} C^T \lambda_{\text{KKT}} \\ &= (X^T X)^{-1} X^T y + \frac{1}{2} (X^T X)^{-1} C^T \cdot 2[C(X^T X)^{-1} C^T]^{-1} [h - C(X^T X)^{-1} X^T y] \\ &= \hat{\beta}_{\text{full}} - (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}_{\text{full}} - h) \end{aligned}$$

其中 $\hat{\beta}_{\text{full}} = (X^T X)^{-1} X^T y$

Part (2)

Find the F-test for testing H_0

(Write the test statistic and prove its null distribution) [10 marks]

Solution:

下面我们计算简约模型误差平方和 $SSE_{\text{reduced}} = \|y - X\hat{\beta}_{\text{reduced}}\|_2^2$

$$\begin{aligned} SSE_{\text{reduced}} &= \|y - X\hat{\beta}_{\text{reduced}}\|_2^2 \\ &= \|y - X(\hat{\beta}_{\text{full}} - (X^T X)^{-1} C^T (C(X^T X)^{-1} C^T)^{-1} (C\hat{\beta}_{\text{full}} - h))\|_2^2 \\ &= \|y - X\hat{\beta}_{\text{full}} + X(X^T X)^{-1} C^T (C(X^T X)^{-1} C^T)^{-1} (C\hat{\beta}_{\text{full}} - h)\|_2^2 \\ &= \|y - X\hat{\beta}_{\text{full}} + A(C\hat{\beta}_{\text{full}} - h)\|_2^2 \quad (\text{denote } A := X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1}) \\ &= \|y - X\hat{\beta}_{\text{full}}\|_2^2 - 2(y - X\hat{\beta}_{\text{full}})^T A(C\hat{\beta}_{\text{full}} - h) + \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2 \end{aligned}$$

其中 $A := X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} \in \mathbb{R}^{n \times m}$.

考虑交叉项 $(y - X\hat{\beta}_{\text{full}})^T A(C\hat{\beta}_{\text{full}} - h)$:

$$\begin{aligned} &(y - X\hat{\beta}_{\text{full}})^T A(C\hat{\beta}_{\text{full}} - h) \\ &= (y - Hy)^T A(C\hat{\beta}_{\text{full}} - h) \quad (\text{recall that } X\hat{\beta}_{\text{full}} = Hy \text{ where } H = X(X^T X)^{-1} X^T y) \\ &= y^T (I_n - H) X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}_{\text{full}} - h) \quad (\text{note that } HX = X \text{ so that } (I_n - H)X = 0_{n \times (p+1)}) \\ &= y^T 0_{n \times (p+1)} (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}_{\text{full}} - h) \\ &= 0 \end{aligned}$$

因此我们有:

$$\begin{aligned} SSE_{\text{reduced}} &= \|y - X\hat{\beta}_{\text{full}}\|_2^2 - 2(y - X\hat{\beta}_{\text{full}})^T A(C\hat{\beta}_{\text{full}} - h) + \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2 \\ &= SSE_{\text{full}} - 2 \cdot 0 + \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2 \\ &= SSE_{\text{full}} + \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2 \end{aligned}$$

我们定义**额外误差平方和** (extra sum of squares, ESS) 为从全模型到简约模型增加的误差平方和:

$$ESS = SSE_{\text{reduced}} - SSE_{\text{full}} = \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2$$

$$\text{其中 } \begin{cases} A = X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} \\ \hat{\beta}_{\text{full}} = (X^T X)^{-1} X^T y \end{cases}$$

考虑第一类型错误概率界限为 α 的检验问题 $H_0 : C\beta = h \leftrightarrow H_1 : C\beta \neq h$

下面我们研究**额外误差平方和** $ESS = SSE_{\text{reduced}} - SSE_{\text{full}} = \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2$ 在零假设 $H_0 : C\beta = h$ 下的分布.

$$\begin{aligned} ESS &= SSE_{\text{reduced}} - SSE_{\text{full}} \\ &= \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2 \\ &= (C\hat{\beta}_{\text{full}} - h)^T A^T A (C\hat{\beta}_{\text{full}} - h) \quad (\text{recall that } A = X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1}) \\ &= (C\hat{\beta}_{\text{full}} - h)^T \{X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1}\}^T \{X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1}\} (C\hat{\beta}_{\text{full}} - h) \\ &= (C\hat{\beta}_{\text{full}} - h)^T \{[C(X^T X)^{-1} C^T]^{-1} C(X^T X)^{-1} X^T\} \cdot \{X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1}\} (C\hat{\beta}_{\text{full}} - h) \\ &= (C\hat{\beta}_{\text{full}} - h)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}_{\text{full}} - h) \\ &= \eta^T \eta \quad (\text{denote } \eta = [C(X^T X)^{-1} C^T]^{-\frac{1}{2}} (C\hat{\beta}_{\text{full}} - h)) \end{aligned}$$

注意到 $\hat{\beta}_{\text{full}} = (X^T X)^{-1} X^T y \sim N(\beta, \sigma^2 (X^T X)^{-1})$

于是我们有:

$$\begin{aligned} C\hat{\beta}_{\text{full}} - h &\sim N(C\beta - h, \sigma^2 C(X^T X)^{-1} C^T) \\ &\stackrel{H_0}{=} N(0_m, \sigma^2 C(X^T X)^{-1} C^T) \quad (\text{where } H_0 : C\beta = h) \end{aligned}$$

因此我们有:

$$\begin{aligned} \eta &= [C(X^T X)^{-1} C^T]^{-\frac{1}{2}} (C\hat{\beta}_{\text{full}} - h) \\ &\stackrel{H_0}{\sim} N([C(X^T X)^{-1} C^T]^{-\frac{1}{2}} \cdot 0_m, [C(X^T X)^{-1} C^T]^{-\frac{1}{2}} \sigma^2 C(X^T X)^{-1} C^T \{[C(X^T X)^{-1} C^T]^{-\frac{1}{2}}\}^T) \quad (\text{where } H_0 : C\beta = h) \\ &= N(0_m, \sigma^2 I_m) \end{aligned}$$

$$\text{于是我们有 ESS} = \eta^T \eta \stackrel{H_0}{\sim} \sigma^2 \chi_{(m)}^2$$

现在我们可以构造线性约束检验问题 $H_0 : C\beta = h \leftrightarrow H_1 : C\beta \neq h$ 的检验统计量了:

$$\begin{aligned} F &:= \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}})/m}{\text{SSE}_{\text{full}}/n - p - 1} \\ &= \frac{\text{ESS}/m}{\text{SSE}_{\text{full}}/n - p - 1} \\ &\quad (\text{note that } \begin{cases} \text{ESS} \stackrel{H_0}{\sim} \sigma^2 \chi_{(m)}^2 \\ \text{SSE}_{\text{full}} \sim \sigma^2 \chi_{(n-p-1)}^2 \end{cases} \text{ where } H_0 : C\beta = h) \\ &\stackrel{H_0}{\sim} \frac{\sigma^2 \chi_{(m)}^2/m}{\sigma^2 \chi_{(n-p-1)}^2/(n-p-1)} \\ &= F_{m, n-p-1} \end{aligned}$$

其中分子 $\frac{1}{m} \text{ESS} = \frac{1}{m} \|A(C\hat{\beta}_{\text{full}} - h)\|_2^2$ 与分母 $s_{\text{full}}^2 = \frac{1}{n-p-1} \|y - X\hat{\beta}_{\text{full}}\|_2^2$ 的独立性由 $\hat{\beta}_{\text{full}} \perp s_{\text{full}}^2$ 保证.
(其中 s_{full}^2 记为全模型中根据 σ^2 的极大似然估计量构造出的无偏估计量)

我们记 $F_{m, n-p-1, \alpha}$ 为 $F_{m, n-p-1}$ 分布的 $1 - \alpha$ 分位数.

则线性约束 $C\beta = h$ 的显著性检验的 F -检验法为:

- (F -检验法)

$$\text{若 } F = \frac{\text{ESS}/m}{\text{SSE}_{\text{full}}/n - p - 1} = \frac{\|A(C\hat{\beta}_{\text{full}} - h)\|_2^2/m}{\|y - X\hat{\beta}_{\text{full}}\|_2^2/(n-p-1)} > F_{m, n-p-1, \alpha}$$

则我们拒绝零假设 $H_0 : C\beta = h$, 即我们认为线性先验关系 $C\beta = h$ 不成立.

$$\text{其中 } \begin{cases} A = X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} \\ \hat{\beta}_{\text{full}} = (X^T X)^{-1} X^T y \end{cases}$$

Problem 6

Consider a linear regression model with Normal errors and take σ as known.

Show that the model with the largeset AIC statistic is the model with the lowest C_p statistic. [10 marks]

Solution:

给定某个 p 元解释变量子集, 设其设计矩阵为 $X_p \in \mathbb{R}^{n \times (p+1)}$

我们记其回归参数向量 β_p 的估计量 $\hat{\beta}_p = (X_p^T X_p)^{-1} X_p^T y$ (这里的下标只代表模型的解释变量个数, 不代表分量)

误差平方和 $\text{SSE}_p = \|y - \hat{y}_p\|_2^2 = \|y - X_p \hat{\beta}_p\|_2^2$ (自由度为 $n - p - 1$)

在正态假设下，我们有 $y \sim N(X_p \beta_p, \sigma^2 I_n)$ (其中 σ^2 反映的是随机噪音的强度，与模型无关)
其概率密度函数为：

$$\begin{aligned} f(y) &= \frac{1}{(\sqrt{2\pi})^n |\sigma^2 I_n|^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}(y - X_p \beta_p)^T (\sigma^2 I_n)^{-1} (y - X_p \beta_p)\right\} \\ &= \frac{1}{(\sqrt{2\pi}\sigma)^n} \exp\left\{-\frac{1}{2\sigma^2} \|y - X_p \beta_p\|_2^2\right\} \end{aligned}$$

似然函数 $L(\beta_p, \sigma^2 | X_p, y)$ 和对数似然函数 $\log L(\beta_p, \sigma^2 | X_p, y)$ 为：

$$\begin{aligned} L(\beta_p, \sigma^2 | X_p, y) &= f(y) = \frac{1}{(\sqrt{2\pi}\sigma)^n} \exp\left\{-\frac{1}{2\sigma^2} \|y - X_p \beta_p\|_2^2\right\} \\ \log L(\beta_p, \sigma^2 | X_p, y) &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \|y - X_p \beta_p\|_2^2 \end{aligned}$$

根据多元线性回归的结论可知，最大化对数似然函数 $\log L(\beta_p, \sigma^2 | X_p, y)$ 得到的似然解为：
(假设 σ^2 已知)

$$\begin{aligned} \max_{\beta_p, \sigma^2} \log L(\beta_p, \sigma^2 | X_p, y) &= \log L(\hat{\beta}_p, \sigma^2 | X_p, y) \quad (\text{where } \hat{\beta}_p = (X_p^T X_p)^{-1} X_p^T y) \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \|y - X_p \hat{\beta}_p\|_2^2 \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \text{SSE}_p \end{aligned}$$

我们希望最大化：

(王勤文老师采用的定义，其中模型复杂度由解释变量个数来表征，而最大对数似然函数舍去了与 p 无关的项)

$$\begin{aligned} \text{AIC}_p &:= \text{maximum_log-likelihood} - \text{model_complexity} \\ &= -\frac{1}{2\sigma^2} \text{SSE}_p - p \end{aligned}$$

在所有回归子集中 Akaike 信息量 AIC_p 最大者对应的回归模型就是最优模型。

在 σ^2 已知的条件下，我们可以构造 C_p 统计量：

$$C_p = \frac{\text{SSE}_p}{\sigma^2} - n + 2(p + 1)$$

在所有回归子集中，我们尽量选择 Mallows 指数 $C_p \approx p + 1$ 的子集。

若有多个子集满足 $C_p \approx p + 1$ (它们对应的 p 可能是不同的)，则我们尽量选取 C_p 值较小的子集。

简单起见，我们认为在所有回归子集中 C_p 统计量最小者对应的回归模型就是最优模型。

因此最大化 AIC_p 和最小化 C_p 所选择的模型是一致的。

The End